

Forward Process (Diffusion Process) Summary

The joint distribution of the forward process is defined as:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (1)$$

The probability distribution of adding noise at a single timestep t :

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad (2)$$

Using the reparameterization trick to generate $\mathbf{x}_t$ from $\mathbf{x}_{t-1}$.

Let $\alpha_t := 1 - \beta_t$ and $\mathbf{z}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1} \quad (3)$$

Similarly, for the previous step $\mathbf{x}_{t-1}$:

$$\mathbf{x}_{t-1} = \sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}}\mathbf{z}_{t-2} \quad (4)$$

Substitute Equation (4) into Equation (3):

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}}\mathbf{z}_{t-2} \right) + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \underbrace{\sqrt{\alpha_t(1 - \alpha_{t-1})}\mathbf{z}_{t-2} + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1}}_{\text{Sum of two independent Gaussian noise terms}} \end{aligned}$$

Since the sum of two independent Gaussian variables $\mathcal{N}(0, \sigma_1^2\mathbf{I}) + \mathcal{N}(0, \sigma_2^2\mathbf{I})$ results in $\mathcal{N}(0, (\sigma_1^2 + \sigma_2^2)\mathbf{I})$, we merge the noise terms. The new variance is:

$$\begin{aligned} \text{Variance} &= \alpha_t(1 - \alpha_{t-1}) + (1 - \alpha_t) \\ &= \alpha_t - \alpha_t\alpha_{t-1} + 1 - \alpha_t \\ &= 1 - \alpha_t\alpha_{t-1} \end{aligned}$$

Thus, merging the noise terms into a new standard Gaussian variable $\mathbf{z}' \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and continuously applying this substitution down to the initial state $\mathbf{x}_0$ with $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$, we obtain the closed-form sampling equation:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon} \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (5)$$

The corresponding marginal probability distribution of reaching $\mathbf{x}_t$ directly from the original data $\mathbf{x}_0$ is:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad (6)$$

Loss Function Design: A Detailed Breakdown

1. From Theoretical Dream to Tractable Loss

Our theoretical goal is to force the neural network (p_θ) to perfectly imitate nature's forward diffusion process (q). This is framed as optimizing the variational upper bound on the negative log-likelihood:

$$\mathbb{E}[-\log p_\theta(\mathbf{x}_0)] \leq \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] =: L \quad (7)$$

The Fatal Flaw: Equation (7) cannot be computed in code because the time directions are mismatched. The neural network p_θ moves backward (from t to $t - 1$), but the denominator q moves forward (from $t - 1$ to t). We cannot directly compare them.

2. The Mathematical Breakthrough: Reversing Time

To solve this time-mismatch, we apply Bayes’ theorem to reverse the time direction of the forward process by conditioning it on the absolute ground truth $\mathbf{x}_0$:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \quad (8)$$

By substituting Equation (8) into Equation (7) and utilizing the properties of logarithms, we can split the summation into two distinct parts:

$$\begin{aligned} L &= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t=1}^T \log \left(\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \cdot q(\mathbf{x}_t|\mathbf{x}_0)} \right) \right] \\ &= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t=1}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} - \underbrace{\sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}}_{\text{The Telescoping Sum}} \right] \end{aligned}$$

The rightmost summation collapses beautifully via telescoping sum. Most terms cancel out with their adjacent counterparts, leaving only:

$$-\log q(\mathbf{x}_0|\mathbf{x}_0) + \log q(\mathbf{x}_T|\mathbf{x}_0) = 0 + \log q(\mathbf{x}_T|\mathbf{x}_0)$$

By reassembling the surviving terms and grouping them for $t = T$, $t > 1$, and $t = 1$, we derive the final tractable loss function:

$$L = \mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right] \quad (9)$$

Now, inside L_{t-1} , both the AI’s prediction (p_θ) and nature’s ground truth (q) move backward from t to $t - 1$. This allows us to calculate their exact difference using Kullback-Leibler (KL) divergence.

3. Fast-Forwarding and the Omniscient Teacher

To compute the KL divergence in Equation (9), we rely on Equation (6) as a “fast-forward” mechanism to generate $\mathbf{x}_t$ directly from $\mathbf{x}_0$.

Furthermore, if we possess a “God’s-eye view”—knowing both the current corrupted image $\mathbf{x}_t$ and the ultimate perfect image $\mathbf{x}_0$ —nature dictates an absolute correct answer for what the previous step $\mathbf{x}_{t-1}$ looked like. This is called the tractable posterior:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \quad (10)$$

This posterior distribution acts as the “Omniscient Teacher”—the absolute standard that the neural network must learn to mimic.

4. The Exact Target Coordinates

Equation (10) defines the shape, but we need its exact coordinates. The true mean (the bullseye, $\tilde{\boldsymbol{\mu}}_t$) and the true variance ($\tilde{\beta}_t$) of the ground truth distribution are explicitly formulated as:

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t \quad \text{and} \quad \tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t \quad (11)$$

The Ultimate Conclusion: The true mean $\tilde{\boldsymbol{\mu}}_t$ is fundamentally a weighted average of the clean image $\mathbf{x}_0$ and the noisy image $\mathbf{x}_t$. The entire training process of a Diffusion Model boils down to forcing the U-Net’s predicted mean to perfectly align with this mathematically derived $\tilde{\boldsymbol{\mu}}_t$.

5. Derivation of the Posterior (The Omniscient Teacher)

We derive the closed form of $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ using Gaussian conditioning.

Step 1: Known distributions from the forward process. From the closed-form marginal and the forward transition:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1 - \bar{\alpha}_{t-1})\mathbf{I}),$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}).$$

Step 2: Joint distribution conditioned on $\mathbf{x}_0$. Since $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the pair $(\mathbf{x}_{t-1}, \mathbf{x}_t)$ is jointly Gaussian given $\mathbf{x}_0$. Their means and covariances are:

$$\begin{aligned}\boldsymbol{\mu}_{t-1} &= \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, & \boldsymbol{\mu}_t &= \sqrt{\alpha_t}\mathbf{x}_0, \\ \Sigma_{t-1,t-1} &= (1 - \bar{\alpha}_{t-1})\mathbf{I}, & \Sigma_{t,t} &= (1 - \bar{\alpha}_t)\mathbf{I}, \\ \Sigma_{t-1,t} &= \text{Cov}(\mathbf{x}_{t-1}, \mathbf{x}_t) = \sqrt{\alpha_t}\Sigma_{t-1,t-1} = \sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{I}.\end{aligned}$$

Step 3: Conditional Gaussian formula. For a jointly Gaussian vector $\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}$, the conditional distribution $\mathbf{a}|\mathbf{b}$ is Gaussian with:

$$\mathbb{E}[\mathbf{a}|\mathbf{b}] = \boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{b} - \boldsymbol{\mu}_b), \quad \text{Cov}(\mathbf{a}|\mathbf{b}) = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}.$$

Here set $\mathbf{a} = \mathbf{x}_{t-1}$, $\mathbf{b} = \mathbf{x}_t$. Then:

$$\tilde{\boldsymbol{\mu}}_t = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_0).$$

Step 4: Simplify the mean. Expand and collect the coefficients of $\mathbf{x}_0$ and $\mathbf{x}_t$:

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_t &= \left(\underbrace{\sqrt{\bar{\alpha}_{t-1}} - \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t}}_{= \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}} \right) \mathbf{x}_0 + \underbrace{\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}}_{= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}} \mathbf{x}_t.\end{aligned}$$

The simplification of the $\mathbf{x}_0$ coefficient uses the identity $1 - \bar{\alpha}_t = \beta_t + \alpha_t(1 - \bar{\alpha}_{t-1})$ and $\sqrt{\bar{\alpha}_t} = \sqrt{\alpha_t\bar{\alpha}_{t-1}}$. The final result is:

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t.$$

Step 5: Conditional variance.

$$\tilde{\beta}_t\mathbf{I} = (1 - \bar{\alpha}_{t-1})\mathbf{I} - \frac{\alpha_t(1 - \bar{\alpha}_{t-1})^2}{1 - \bar{\alpha}_t}\mathbf{I} = \frac{(1 - \bar{\alpha}_{t-1})\beta_t}{1 - \bar{\alpha}_t}\mathbf{I}.$$

Hence $\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$.

Therefore,

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t\mathbf{I}),$$

which matches Equation (11).